Characteristic bounds for the past scalar response of a timelike shell

Ricardo Maldonado
Independent Researcher
Henderson, Nevada, United States

4 November 2026 (America/Los_Angeles)

Abstract

For one registered packet in a conditional linear Einstein–scalar shell construction, the past–future difference has zero initial displacement and a velocity input with small integrated mass. Incoming-characteristic transport, positive two-variable majorants and a boundary-inertia comparator bound the actual physical scalar-coefficient difference by $1/40$ on $0 \le t \le 3\ell$, and by $1/40000$ on $49\ell/25 \le t \le 3\ell$, where $\ell = 10^{-6}$. The full dynamic boundary law and the complete previous forcing history are retained. An explicit normalization join to older actual future enclosures transfers two disjoint fine zero brackets and a strictly negative post-pulse coefficient tail to the past. A separate norm/cross-term estimate also bounds the past first-field quadratic scalar norm. The older analytic and waveform-certificate premises remain explicit; their full campaigns are not rerun. These are first-coefficient results in model units, without an effective nonlinear amplitude interval or a cosmological interpretation.

1 Registered problem and results

Fix $\ell = 10^{-6}$ and $T = 3\ell$. Set

$$Q(x) = \begin{cases} \exp\!\left(4 - \dfrac{1}{(x-1)(2-x)}\right), & 1 < x < 2, \\ 0, & \text{otherwise}, \end{cases} \qquad q(z) = Q(z/\ell), \quad h(z) = e^{-z} q(z).$$

The registered current master has data $(w, w_t)|_{t=0} = (q, h)$ and solves the complete reduced problem retained from the contact construction [1]:

$$\begin{aligned} w_{tt} - w_{zz} + W(t,z)w = 0, \qquad & -w_z + \mu w_{tt} + \beta(t) w = 0 \quad (z = 0), \\ W(t,z) = V(z) - a_0 \tanh^2 t, \qquad & \beta(t) = \beta_0 - a_0 \mu \tanh^2 t, \\ \beta_0 = \gamma + \tfrac{5}{2}\mu, \qquad & a_0 = \tfrac{35}{4}. \end{aligned} \tag{1}$$

Its embedding-included first physical scalar coefficient is

$$\begin{aligned} \Phi_1(\tau, \Omega) = S(\tau/\rho_b) Y(\Omega), \qquad & Y = n_1^2 - n_2^2, \quad \tau = \rho_b t, \\ S(t) = G \cosh^{-3/2} t \, [w_z(t,0) - \gamma w(t,0)], \qquad & G = \frac{3}{\rho_b^3 p_b}. \end{aligned} \tag{2}$$

The background shell scalar is constant, so first clock/label pullbacks add no amplitude term. The angular mean of Y is zero and $\sup |Y| = 1$. The branch, reconstruction, regularity, full boundary law and smooth linear development/uniqueness are retained premises. The older actual future waveform theorem [2] is additionally retained when its intervals are used below. Exact selected-file checks do not reprove these premises.

Theorem 1. *Under these premises, the registered first scalar amplitude obeys*

$$\sup_{0\le t\le 3\ell} |S(-t) - S(t)| < \frac{1}{40}, \tag{3}$$

$$\sup_{49\ell/25\le t\le 3\ell} |S(-t) - S(t)| < \frac{1}{40000}. \tag{4}$$

The paired proper times are $-\rho_b t$ and $+\rho_b t$ at one anchor. The bounds concern absolute differences of first linear coefficients in model units, not either individual amplitude or a relative error.

2 Zero displacement and complete boundary majorants

Let w_+ and w_- have data $(q, +h)$ and $(q, -h)$ for the same problem (1). Evenness of both W and β, followed by uniqueness, gives $w_{\mathrm{current}}(-t) = w_-(t)$ and $w_{\mathrm{current}}(t) = w_+(t)$. Thus

$$\psi = \tfrac{1}{2}(w_+ - w_-), \qquad (\psi, \psi_t)|_0 = (0, h),$$
$$S(-t) - S(t) = -2G\cosh^{-3/2} t\, D_\psi(t), \qquad D_\psi = \psi_z(t,0) - \gamma\psi(t,0).$$

The spatial normal does not reverse. The factor two is essential. This is an actual coupled velocity-only solution, rather than a free approximation or a second-order preparation difference. Its initial shell jets vanish, and its shell trace vanishes before $t = \ell$.

The selected contact enclosures give safe bounds on the relevant causal triangles:

$$\begin{aligned} &\mu > m = \frac{1}{1000}, && |W| < M = 103000, \\ &0 < \beta_0 < B = 300, && |\beta| < B, \quad |\gamma| < \Gamma = 300, \\ &|G| \le G_* := \frac{25000}{129^3} < \frac{3}{250}. \end{aligned} \tag{5}$$

These bounds follow from the admitted shell and background data. Positive elapsed time is supplied by the complete even problem; the older past-only interval is not silently enlarged.

Write $y(t) = \psi(t, 0)$. The velocity is nonnegative, with $\int_0^\infty h\,\mathrm{d}z \le L := \ell$. Its free Dirichlet solution u_D has $|u_D| \le L/2 \le L$. With the half-line retarded kernel $\mathcal{R}$,

$$\psi(t, z) = u_D(t, z) + \mathbf{1}_{t\ge z} y(t - z) + \mathcal{R}[-W\psi](t, z). \tag{6}$$

The kernel is positive and has total mass at most $t^2/2$. On $t + z \le T$, put $A = \sup|\psi|$ and $V_b = \sup_{[0,T]}|y'|$. Incoming transport is exact:

$$(\psi_t + \psi_z)(t, 0) = h(t) - I(t), \qquad I(t) = \int_0^t W(s, t-s)\psi(s, t-s)\,\mathrm{d}s, \qquad \mu y'' + y' = h - I - \beta y. \tag{7}$$

Using $|I(t)| \le MtA$, $|y(t)| \le tV_b$, and the positive exponential kernel of the last equation yields the two inequalities

$$(1 - r)A - TV_b \le L, \qquad -\frac{r}{m}A + (1 - s)V_b \le \frac{L}{m}, \qquad r = \frac{MT^2}{2}, \quad s = \frac{BT^2}{2m}.$$

Their inverse is positive because $d = (1 - r)(1 - s) - Tr/m > 0$. Therefore

$$A \le A_* := \frac{L(1 - s + T/m)}{d}, \qquad V_b \le V_* := \frac{L}{md}. \tag{8}$$

The triangles and their characteristic subdomains stay inside the retained collar. No field or coefficient oracle is sampled. The improvement comes from the velocity mass L, not from treating the original displacement as small.

3 An actual normal-trace estimate

On an incoming ray, odd extension of h gives

$$u_D(s,t-s)=\tfrac12\int_{t-2s}^{t}h_{\mathrm{odd}}(r)\,\mathrm{d}r\ge 0,\qquad \int_0^t u_D(s,t-s)\,\mathrm{d}s=\tfrac12\int_0^t rh(r)\,\mathrm{d}r\le\frac{3\ell^2}{4}.$$

The boundary lift and the bulk remainder in (6) consequently give, on the complete interval,

$$|I(t)|\le I_*:=\frac34M\ell^2+\frac14MV_*T^2+\frac16M^2A_*T^3. \tag{9}$$

In particular, there is no displacement contribution of order $Mtq(t)$.

Let u solve the free bulk equation with the same data $(0,h)$ and constant complete boundary coefficient β_0. Its value $b(t)=u(t,0)$ satisfies $\mu b''+b'+\beta_0 b=h$, with zero initial data. For $q_e=y-b$ and $v_e=q_e'$, subtraction gives

$$\mu v_e'+v_e+\beta_0 q_e=-I+a_0\mu\tanh^2 t\,y.$$

Keeping the whole earlier forcing history, rather than substituting a tail estimate into an on-pulse convolution, gives

$$|v_e|\le E_*:=\frac{TI_*/m+a_0V_*T^4/4}{1-s},\qquad |q_e|\le TE_*,\qquad |D_\psi-D_u|\le R_*:=I_*+(1+\Gamma T)E_*. \tag{10}$$

Indeed $D_\psi-D_u=-I-v_e-\gamma q_e$ by incoming transport. This is a characteristic estimate of the actual normal trace; it is not an inference from an L^2 bulk-energy norm.

For the impulse kernel k, with $\mu k''+k'+\beta_0k=0$, $k(0)=0$, $k'(0)=1/\mu$, the identity

$$[\mu(k')^2+\beta_0k^2]'=-2(k')^2$$

gives $|k'|\le 1/m$ and $|k(t)|\le t/m$. Thus

$$D_u=h-(k'+\gamma k)*h,\qquad |(k'+\gamma k)*h|\le C_{\mathrm{mem}}:=(1+\Gamma T)\ell/m.$$

Here $*$ denotes convolution in nonnegative time. Combining this with (10) proves the uniform estimate

$$|S(-t)-S(t)|\le 2G_*[h(t)+C_{\mathrm{mem}}+R_*]. \tag{11}$$

Exact rational evaluation with $h\le 1$ makes the right side less than $1/40$. For $x\ge 49/25$, the bump decreases to its flat endpoint and

$$Q(49/25)=e^{-529/24}<e^{-22}<10^{-9}.$$

The last inequality follows from the positive series bound $e>8/3$ and the exact integer inequality $(8/3)^{22}>10^9$. Beyond $x=2$, $Q=0$. Replacing h by 10^{-9} in (11) makes its bound less than $1/40000$, proving Theorem 1. An independent route estimates the incoming trace directly from a complete two-variable majorant; it obtains the same two safe ceilings without a comparator oracle.

4 Two past zeros, a signed tail and a quadratic norm

The retained wider future certificate uses this same $\ell=10^{-6}$ packet, full dynamic boundary law, physical normal and anchored clock. Its actual enclosures include the error from its free comparator. The exported `ell_S` is ℓS, so division by ℓ precedes widening. The distinct $\ell=10^{-10}$ state or a comparator-only value cannot replace this actual-field input. Exact interval widening gives Table 1.

Matched $x = t/\ell$	Difference ceiling	Strict past coefficient bound
1.49998	1/40	$S(-\ell x) < -3/2$
1.49999	1/40	$S(-\ell x) > 1$
1.96	1/40000	$S(-\ell x) > 1/1000$
1.97	1/40000	$S(-\ell x) < -29/100000$

Table 1: Safe rational sign margins after converting the retained actual future intervals to physical-S model units. The terminating time decimals are exact rational numbers.

Corollary 2. *The continuous first scalar amplitude has at least one zero in each of the two disjoint open past-coordinate intervals*

$$(-1.49999\ell, -1.49998\ell), \qquad (-1.97\ell, -1.96\ell). \tag{12}$$

Its post-pulse past coefficient satisfies uniformly

$$-\frac{1}{1500} < S(-t) < -\frac{1}{4000}, \qquad 2\ell \le t \le 3\ell. \tag{13}$$

The zeros follow from the strict signs and continuity. The tail follows from the retained whole-interval actual future tail enclosure and the localized bound (4). These are at least two distinct zeros, without uniqueness, a complete zero count or optimality. The central width is $\ell/100000$, a factor $3125/64$ narrower than the preceding past $\ell/2048$ bracket [3]. This transfers an older fine future bracket; it does not improve that future bracket. A negative S is not a globally negative angular field because Y changes sign.

For completeness a separate norm estimate retains the quadratic cross term. Define $H_\pm(x) = \ell D_{\mathrm{current}}(\pm\ell x)$, $I_\pm = \int_0^3 H_\pm^2 \,\mathrm{d}x$. From the characteristic bound, the prefront vanishing and $\Delta H = -2\ell D_\psi(\ell x)$,

$$\|\Delta H\|^2_{L^2(0,3)} \le 2[2\ell(1 + C_{\mathrm{mem}} + R_*)]^2 < (3\ell)^2.$$

The retained actual future interval gives $I_+ < 9$, hence

$$|I_- - I_+| \le 2\|H_+\|_2\|\Delta H\|_2 + \|\Delta H\|_2^2 < 18\ell + 9\ell^2 = \frac{18000009}{10^{12}}. \tag{14}$$

For the first-field squared scalar norm on the past background proper-time window, the spatial measure is $\mathrm{d}V_{0,\tau} = \rho_b^3 \cosh^3 t \,\mathrm{d}\Omega$, $\mathrm{d}\tau = \rho_b \,\mathrm{d}t$, and $\int_{S^3} Y^2 \,\mathrm{d}\Omega = \pi^2/3$. The even background measure and conjugation cancel as in the future:

$$\mathcal{C}_- = \frac{\mathcal{K}}{\ell} I_-, \qquad \mathcal{K} = \frac{3\pi^2}{\rho_b^2 p_b^2} > 0, \qquad 6.06\,10^{11} < \mathcal{C}_- < 7.44\,10^{11}. \tag{15}$$

The time reversal changes integration limits, not the positive sign of this integral. The last interval follows by widening the retained actual I_+ interval with (14) and multiplying the retained positive $\mathcal{K}/\ell$ interval. This is not physical injected or Noether energy.

Scope and reproducibility

The new calculations use selected public input copies, continuous proofs, independent derivations and exact normal/optimized arithmetic. No old waveform campaign, numerical PDE, background trajectory or field/source/bump oracle is rerun. Authentication and finite arithmetic do not replace the retained analytic and actual-waveform theorems. The source capsule includes a separate manuscript display audit. Internal review is distinct from external peer review and proof-assistant certification.

Earlier project results already include conditional formal complete future second Taylor coefficients and conditional exact nonlinear future families, with qualitative small-amplitude conclusions [4, 5, 6]. Those results are preserved. Applying the appropriate hypotheses to a chosen past completion, quantifying its particular observable and remainder, and obtaining an effective parameter radius remain distinct tasks. No common exact past family or finite-amplitude waveform is established in this note. The bounds provide neither SI time calibration, a physical source mechanism, radiation/heating nor evidence for a higher-dimensional Big Bang.

The methods have direct characteristic and positive-majorant precedents in the older project proof [2]; they are not new general principles. Recent timelike-Einstein boundary work distinguishes reduced existence, constraint propagation and geometric consistency [7]. Its vacuum, dimension and boundary hypotheses do not supply this shell problem. Similarly, damped Wentzell stabilization [8] is not imported as a backward decay theorem. The comparison is focused, not an exhaustive priority review; broader novelty and significance require specialist assessment.

References

[1] R. Maldonado, *A pointwise scalar-trace bound through past shell contact*, working paper and source capsule (4 November 2026), HDBLAST past-shell-contact checkpoint.

[2] R. Maldonado, *A wider shell waveform with a certified signed inertial tail*, primary proof and actual enclosure report (15 October 2026), HDBLAST inertial-waveform checkpoint.

[3] R. Maldonado, *A past–future scalar comparison for a timelike-shell packet*, working note and source capsule (4 November 2026), HDBLAST past–future scalar-transfer checkpoint.

[4] R. Maldonado, *A complete finite-angular second-order coefficient through shell contact* (14 October 2026), full-contact primary proof.

[5] R. Maldonado, *Exact finite-regularity shell evolution* (15 October 2026), HDBLAST quasilinear-shell-evolution primary proof.

[6] R. Maldonado, *Intrinsic shell budget*, derivation (1 November 2026), sections on anchored proper-clock cuts and family-wise small-amplitude sign transfer.

[7] C. Stevens and J. A. Valiente Kroon, *Wellposedness of the initial boundary value problem for the conformal field equations*, arXiv:2604.24931v1 (2026), sections 5.2–7.

[8] H. Parada and N. Vanspranghe, *Optimal stabilization rate for the wave equation with hyperbolic boundary condition*, arXiv:2512.19167v1 (2025), section 1.1.